

Module 13 Assignment - Arnav Kucheriya

Friday, April 17, 2026

Q1) Advantages of Segmentation over Paging

1. Logical Program Structure

Segmentation reflects how programs are naturally divided (code, data, stack), making memory easier to manage.

2. Better Protection and Sharing

Each segment can have different access permissions, improving security and allowing controlled sharing.

Q2) Logical to Physical Address Translation

Given:

- Maximum segment size = $1K = (2^{10}) \rightarrow$ Offset = 10 bits
- Logical address = 16 bits $\rightarrow$ Segment = 6 bits
- Logical address: 0011100010111010

Step 1: Separate Segment and Offset

- Segment number (first 6 bits):
 $001110_2 = 14_{10}$
- Offset (last 10 bits):
 $0010111010_2 = 186_{10}$

Step 2: Compute Base Address

Base Address = $(2^2 + 4096 + \{\text{Segment No.}\})$
 $= 4 + 4096 + 14$
 $= 4114$

Step 3: Compute Physical Address

Physical Address = Base + Offset
 $= 4114 + 186$
 $= 4300$

Step 4: Convert to Binary

$4300_{10} = 1000011001100_2$

Final Answer:

Physical Address = 4300 (decimal)

Binary = 1000011001100

Q3) FIFO Page Replacement

Reference String:

1, 2, 3, 0, 1, 2, 4, 1, 2, 3, 0, 4, 2, 1, 2, 3, 0, 4

Frames = 4 (initially empty)

Final Results:

- Page Faults = 15
- Page Hits = 3

- Page Fault Ratio = $15 / 18 = 0.833$
- Hit Ratio = $3 / 18 = 0.167$

Q4) LRU Page Replacement

Final Results:

- Page Faults = 12
- Page Hits = 6
- Page Fault Ratio = $12 / 18 = 0.667$
- Hit Ratio = $6 / 18 = 0.333$

Q5) Optimal Page Replacement

Final Results:

- Page Faults = 7
- Page Hits = 11
- Page Fault Ratio = $7 / 18 \approx 0.389$
- Hit Ratio = $11 / 18 \approx 0.611$

Summary Table

Algorithm: FIFO

- Page Faults: 15
- Page Hits: 3
- Fault Ratio: 0.833
- Hit Ratio: 0.167

Algorithm: LRU

- Page Faults: 12
- Page Hits: 6
- Fault Ratio: 0.667
- Hit Ratio: 0.333

Algorithm: Optimal

- Page Faults: 7
- Page Hits: 11
- Fault Ratio: 0.389
- Hit Ratio: 0.611